



Documentation of completed project

2025/2026

High Availability Distributed Systems

Project 1

Field of study: Computer Science

Team members:

Mateusz Fydrych s193410

Adam Chabraszewski s193373

Gdańsk, 2025/2026

Contents

1	Introduction	2
1.1	Roles in the implementation of the project	2
1.2	The purpose of the project	2
2	Assumptions for the implementation of the project	3
2.1	Technical and non-technical assumptions	3
2.2	Technology stack	3
2.3	Expected results of the project implementation	4
3	Implementation of the project	5
3.1	IP addressing plan	5
3.2	Setting up devices	7
3.2.1	Router configuration	7
3.2.2	PC and server configuration	10
3.3	Routing	12
3.3.1	Decision making	12
3.3.2	Static routing configuration	12
3.4	Network without ACL	14
3.4.1	HTTP and FTP server	14
3.4.2	Tests without ACL	15
3.5	Access Control Lists	17
3.5.1	ACL configuration for HR	18
3.5.2	ACL configuration for Accounting	18
3.5.3	ACL configuration for Development	18
3.5.4	Explanation of selected ACL commands	19
3.5.5	Connectivity tests after ACL	20
3.6	IoT devices	21
4	Cost estimate	26
5	Conclusions	27
5.1	Observations	27
5.2	Achievements	27
5.3	Development potential	27

1 Introduction

1.1 Roles in the implementation of the project

The project was prepared by a two person team. Adam was mainly responsible for preparing the physical topology in Cisco Packet Tracer and creating the IP addressing plan. Mateusz configured routing, access control lists and IoT devices.

The final testing was done together because after every bigger change we had to check if the network still worked correctly. The report was also prepared together, using screenshots and test results from the Packet Tracer project.

1.2 The purpose of the project

The purpose of this project was to design and configure a small enterprise intranet network in Cisco Packet Tracer. The company consists of several parts: HR, Accounting, Software Development, the President office and the Secretary office.

The main requirement was to give all users access to internal web and FTP services, while keeping ordinary departments separated from each other. HR, Accounting and Software Development should only communicate inside their own LAN and with the server. The President and the Secretary should have access to all computers. Additionally, the Secretary should be able to access devices representing the company gate and the lawn watering system.

The final network includes:

- separate LAN segments for each department,
- access routers separating departments,
- a dedicated intranet network connecting routers and the server,
- static routing between routers,
- HTTP and FTP services,
- access control lists used to block unauthorized traffic,
- IoT devices connected to the Secretary LAN.

2 Assumptions for the implementation of the project

2.1 Technical and non-technical assumptions

The project was based on the following assumptions:

1. The company has 10 office workers:
 - 1 employee in HR,
 - 2 employees in Accounting,
 - 7 employees in Software Development.
2. The President has one computer in his office.
3. The Secretary has one computer in the adjacent office.
4. Every department is placed in a separate LAN.
5. Access routers are used to separate LAN segments.
6. All routers and the server are connected through one dedicated intranet network.
7. All users should have access to HTTP and FTP services.
8. Ordinary departments should not have access to other departments.
9. The President and the Secretary should have access to all computers.
10. The Secretary should have access to the gate and lawn watering system.
11. The IP addressing scheme can be freely defined inside the private enterprise network.

Private IPv4 addressing was used in the project. The addressing scheme was selected to be simple and readable. Each department received a separate /24 subnet.

2.2 Technology stack

The tools and technologies used in the project are listed in Table 1.

Technology / tool	Purpose in the project
Cisco Packet Tracer	Network design, simulation, configuration and testing.
Cisco 2911 routers	Access routers used to separate LAN segments.
Cisco 2960 switches	Layer 2 switching in local LANs and in the intranet segment.
IPv4	Addressing of routers, computers, server and IoT devices.
Static routing	Manual routing between departmental networks.
Access Control Lists	Blocking unauthorized communication between departments.
HTTP	Web service available in the company intranet.
FTP	File service available in the company intranet.
ICMP ping	Basic connectivity testing.
IoT devices	Garage Door and Lawn Sprinkler controlled from the Secretary LAN.

Table 1: Technology stack

2.3 Expected results of the project implementation

After completing the configuration, the network should work according to the access rules presented in Table 2.

Source	Destination	Expected result
HR	Web/FTP server	Allowed
Accounting	Web/FTP server	Allowed
Software Development	Web/FTP server	Allowed
President	All computers and server	Allowed
Secretary	All computers, server and IoT devices	Allowed
HR	Accounting, Development, President and Secretary LANs	Denied
Accounting	HR, Development, President and Secretary LANs	Denied
Software Development	HR, Accounting, President and Secretary LANs	Denied

Table 2: Expected access policy

3 Implementation of the project

This section describes the main stages of the implementation: topology design, addressing, device configuration, routing, server configuration, ACL configuration, IoT configuration and final testing.

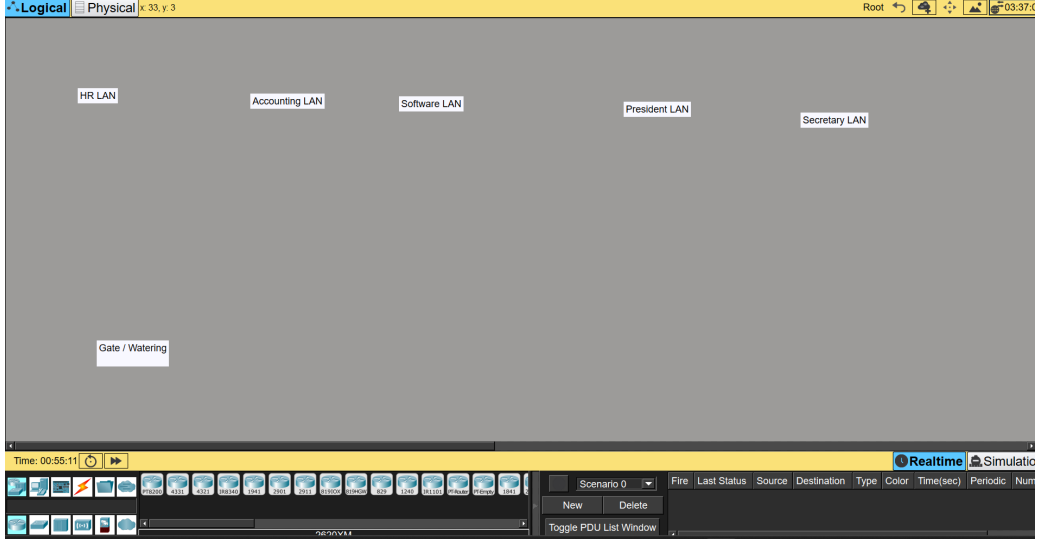


Figure 1: Initial idea of the enterprise network structure

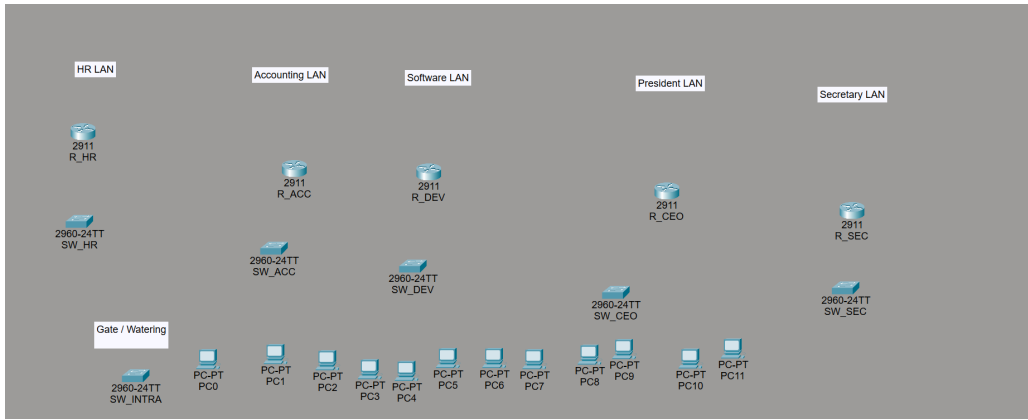


Figure 2: Routers, switches and PCs added to the Packet Tracer workspace

At the beginning, the general logical structure of the network was prepared. After that, the IP addressing plan was created. Each department was assigned a separate /24 subnet to make configuration and troubleshooting easier.

3.1 IP addressing plan

The main addressing plan is shown in Table 3.

Segment	Network	Mask	Default gateway
HR LAN	192.168.10.0	255.255.255.0	192.168.10.1
Accounting LAN	192.168.20.0	255.255.255.0	192.168.20.1
Development LAN	192.168.30.0	255.255.255.0	192.168.30.1
President LAN	192.168.40.0	255.255.255.0	192.168.40.1
Secretary LAN	192.168.50.0	255.255.255.0	192.168.50.1
Intranet	10.0.0.0	255.255.255.0	—

Table 3: IP addressing plan

The addressing of end devices is presented in Table 4.

Device	IP address	Subnet mask	Default gateway
PC_HR	192.168.10.10	255.255.255.0	192.168.10.1
PC_ACC1	192.168.20.10	255.255.255.0	192.168.20.1
PC_ACC2	192.168.20.11	255.255.255.0	192.168.20.1
PC_DEV1	192.168.30.10	255.255.255.0	192.168.30.1
PC_DEV2	192.168.30.11	255.255.255.0	192.168.30.1
PC_DEV3	192.168.30.12	255.255.255.0	192.168.30.1
PC_DEV4	192.168.30.13	255.255.255.0	192.168.30.1
PC_DEV5	192.168.30.14	255.255.255.0	192.168.30.1
PC_DEV6	192.168.30.15	255.255.255.0	192.168.30.1
PC_DEV7	192.168.30.16	255.255.255.0	192.168.30.1
PC_CEO	192.168.40.10	255.255.255.0	192.168.40.1
PC_SEC	192.168.50.10	255.255.255.0	192.168.50.1
Garage Door	192.168.50.20	255.255.255.0	192.168.50.1
Lawn Sprinkler	192.168.50.21	255.255.255.0	192.168.50.1
Web/FTP Server	10.0.0.10	255.255.255.0	10.0.0.1

Table 4: End device addressing

Router interface addressing is shown in Table 5.

Router	Interface	Purpose	IP address
R_HR	G0/0	HR LAN	192.168.10.1
R_HR	G0/1	Intranet	10.0.0.1
R_ACC	G0/0	Accounting LAN	192.168.20.1
R_ACC	G0/1	Intranet	10.0.0.2
R_DEV	G0/0	Development LAN	192.168.30.1
R_DEV	G0/1	Intranet	10.0.0.3
R_CEO	G0/0	President LAN	192.168.40.1
R_CEO	G0/1	Intranet	10.0.0.4
R_SEC	G0/0	Secretary LAN	192.168.50.1
R_SEC	G0/1	Intranet	10.0.0.5

Table 5: Router interface addressing

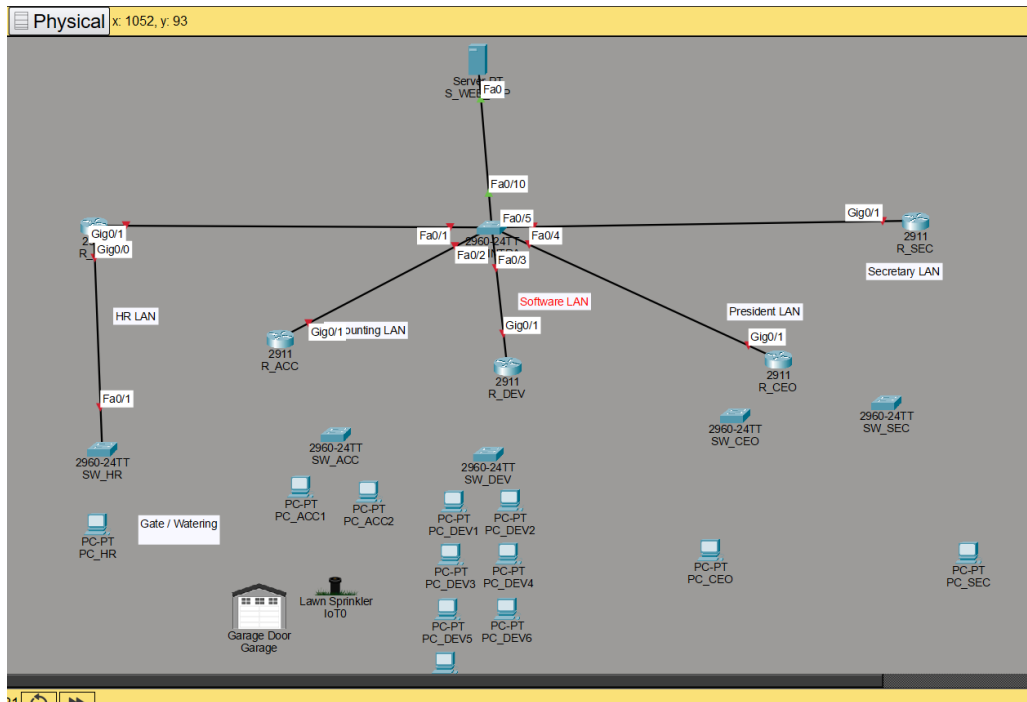


Figure 3: Connecting devices using Copper Straight-Through cables

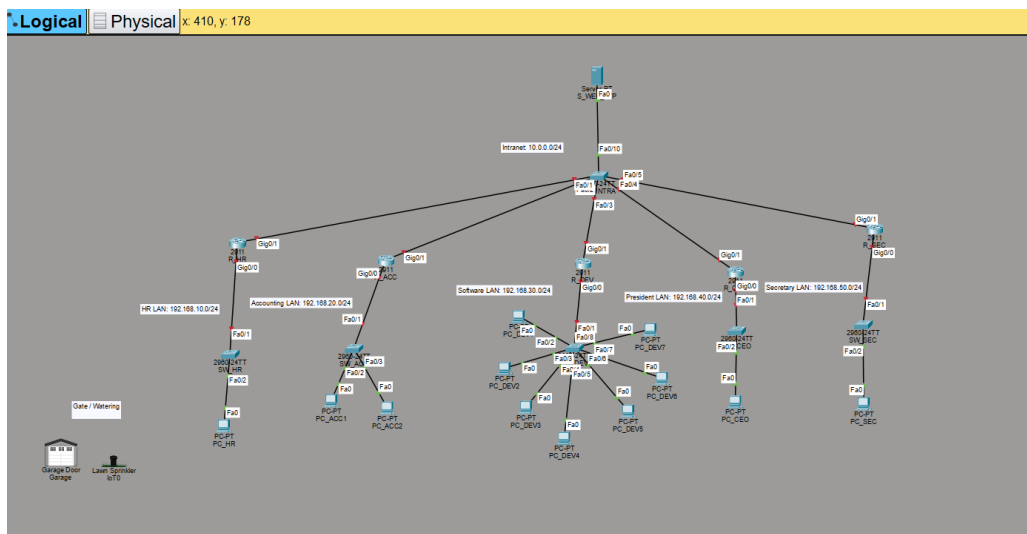


Figure 4: Finished logical architecture

In this topology, every router uses the G0/0 interface for the local LAN and the G0/1 interface for the intranet network.

3.2 Setting up devices

3.2.1 Router configuration

Before logical configuration, the devices did not have assigned IP addresses. This can be seen in the initial `ipconfig` and `show ip interface brief` outputs.

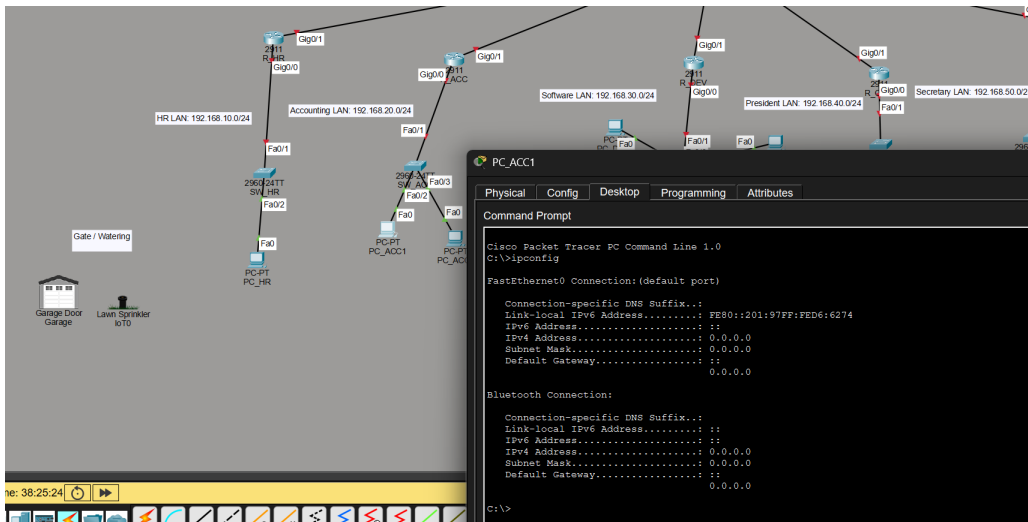


Figure 5: Initial ipconfig output before PC addressing

```

R_HR
Physical Config CLI Attributes
IOS Command Line Interface

http://www.cisco.com/ww1/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>show ip interface brief
Interface      IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0  unassigned      YES unset    administratively down down
GigabitEthernet0/1  unassigned      YES unset    administratively down down
GigabitEthernet0/2  unassigned      YES unset    administratively down down
Vlan1          unassigned      YES unset    administratively down down
Router>

```

Figure 6: Initial show ip interface brief output before router configuration

The routers were configured manually. Static addressing was chosen because the network is small and the number of devices is known in advance. This also makes the configuration easier to check and document.

Example configuration of the HR router:

```

enable
configure terminal
hostname R_HR

```

```

interface gigabitEthernet0/0
description HR_LAN
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
description INTRANET
ip address 10.0.0.1 255.255.255.0
no shutdown
exit

end
write memory

```

Example configuration of the Secretary router:

```

enable
configure terminal
hostname R_SEC

interface gigabitEthernet0/0
description SECRETARY_LAN
ip address 192.168.50.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
description INTRANET
ip address 10.0.0.5 255.255.255.0
no shutdown
exit

end
write memory

```

The same pattern was used for the remaining routers. Only the IP addresses were changed according to the addressing plan.

After configuration, the `show ip interface brief` command was used to verify whether the interfaces were active.

```

R_HR>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.10.1    YES manual up          up
GigabitEthernet0/1       10.0.0.1        YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R_HR>

```

Figure 7: Router interfaces after configuration

Example verification output for the remaining routers:

```

R_ACC#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.20.1	YES	manual	up
GigabitEthernet0/1	10.0.0.2	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down
R_DEV#show ip interface brief				
Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.30.1	YES	manual	up
GigabitEthernet0/1	10.0.0.3	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down
R_CEO#show ip interface brief				
Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.40.1	YES	manual	up
GigabitEthernet0/1	10.0.0.4	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down
R_SEC#show ip interface brief				
Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.50.1	YES	manual	up
GigabitEthernet0/1	10.0.0.5	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

3.2.2 PC and server configuration

Each PC was configured with:

- an IPv4 address,
- a subnet mask,
- a default gateway,
- a DNS server address.

The DNS server address was set to 10.0.0.10, which is the address of the internal server. DNS was not required for basic ping tests, but it was added for completeness and possible future extension.

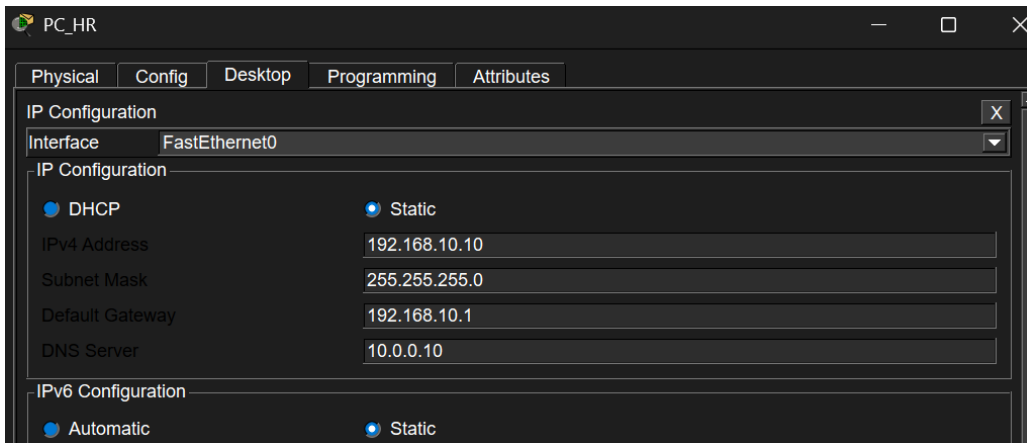


Figure 8: Correctly configured HR PC

For example, the HR PC was configured as follows:

- IP address: 192.168.10.10,
- subnet mask: 255.255.255.0,
- default gateway: 192.168.10.1,
- DNS server: 10.0.0.10.

As a reminder, it is important to note that: the IP address identifies a host inside the network. The subnet mask tells the device which part of the address is the network part and which part is the host part. If the destination address is outside the local subnet, the packet is sent to the default gateway.

The Web/FTP server was placed in the intranet network and configured with the address 10.0.0.10/24. Its default gateway was set to 10.0.0.1, which is the intranet interface of the HR router.

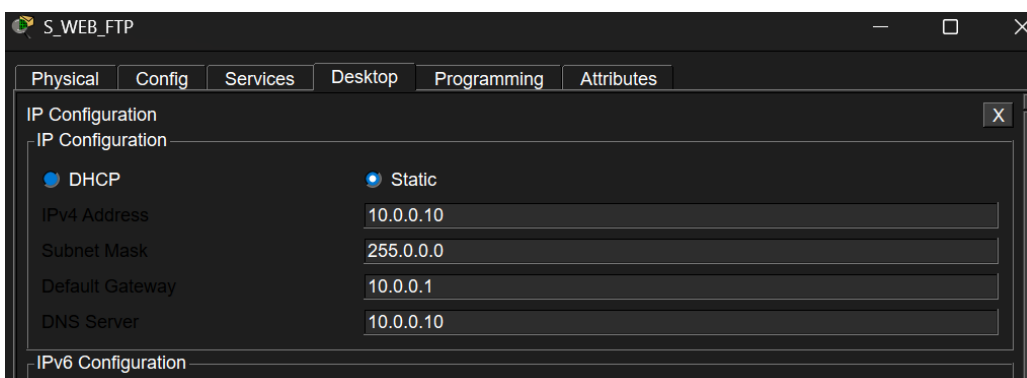


Figure 9: Correctly configured Web/FTP server

Before routing was configured, computers could communicate only inside their own local LAN. This is possible because devices in the same LAN are connected through a switch and can communicate using MAC addresses. Communication with another subnet requires a router and a proper routing table.

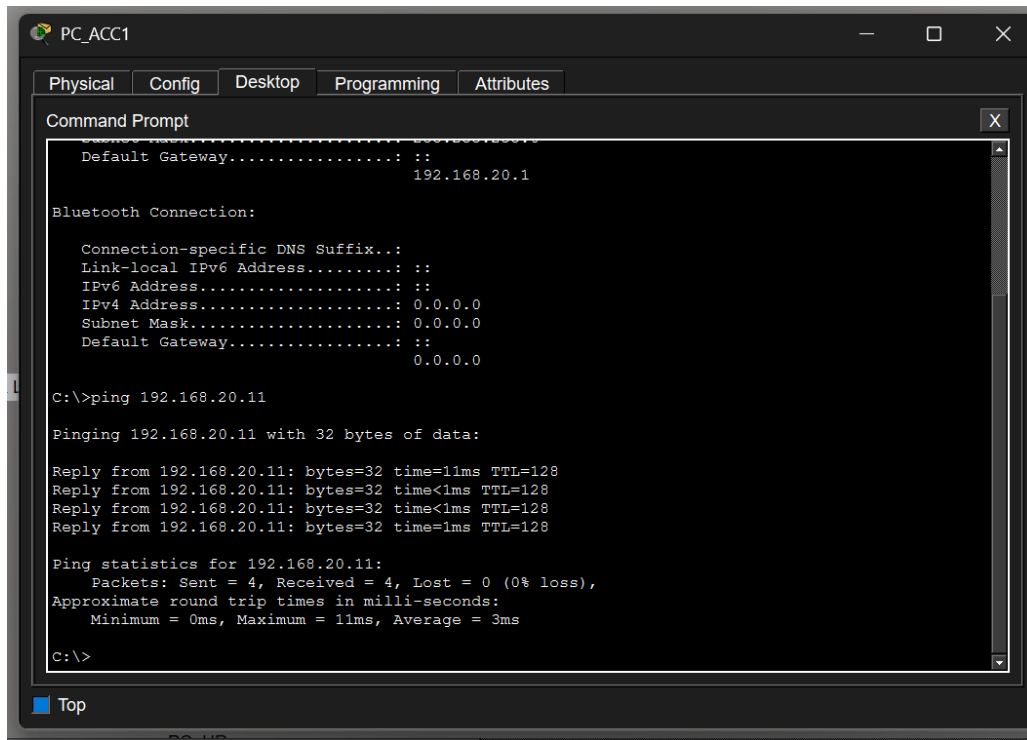


Figure 10: Successful ping test inside the same LAN before routing configuration

3.3 Routing

3.3.1 Decision making

Routing is needed when packets must be sent between different IP networks. In this project, routing is required because every department has its own subnet.

There are two common routing approaches:

- static routing,
- dynamic routing protocols, such as RIP, OSPF or EIGRP.

Dynamic routing is useful in larger networks because routers can exchange route information automatically. In this project, the topology is small and stable, so static routing is enough. It is also easier to understand and verify.

3.3.2 Static routing configuration

All routers are connected to the central intranet network 10.0.0.0/24. Therefore, each router needs static routes pointing to the intranet IP address of the router responsible for the destination subnet.

For example, the following routes were added on the HR router:

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2
ip route 192.168.30.0 255.255.255.0 10.0.0.3
ip route 192.168.40.0 255.255.255.0 10.0.0.4
ip route 192.168.50.0 255.255.255.0 10.0.0.5
```

The complete static routing plan is shown in Table 6.

Router	Destination network	Next hop
R_HR	192.168.20.0/24	10.0.0.2
R_HR	192.168.30.0/24	10.0.0.3
R_HR	192.168.40.0/24	10.0.0.4
R_HR	192.168.50.0/24	10.0.0.5
R_ACC	192.168.10.0/24	10.0.0.1
R_ACC	192.168.30.0/24	10.0.0.3
R_ACC	192.168.40.0/24	10.0.0.4
R_ACC	192.168.50.0/24	10.0.0.5
R_DEV	192.168.10.0/24	10.0.0.1
R_DEV	192.168.20.0/24	10.0.0.2
R_DEV	192.168.40.0/24	10.0.0.4
R_DEV	192.168.50.0/24	10.0.0.5
R_CEO	192.168.10.0/24	10.0.0.1
R_CEO	192.168.20.0/24	10.0.0.2
R_CEO	192.168.30.0/24	10.0.0.3
R_CEO	192.168.50.0/24	10.0.0.5
R_SEC	192.168.10.0/24	10.0.0.1
R_SEC	192.168.20.0/24	10.0.0.2
R_SEC	192.168.30.0/24	10.0.0.3
R_SEC	192.168.40.0/24	10.0.0.4

Table 6: Static routing plan

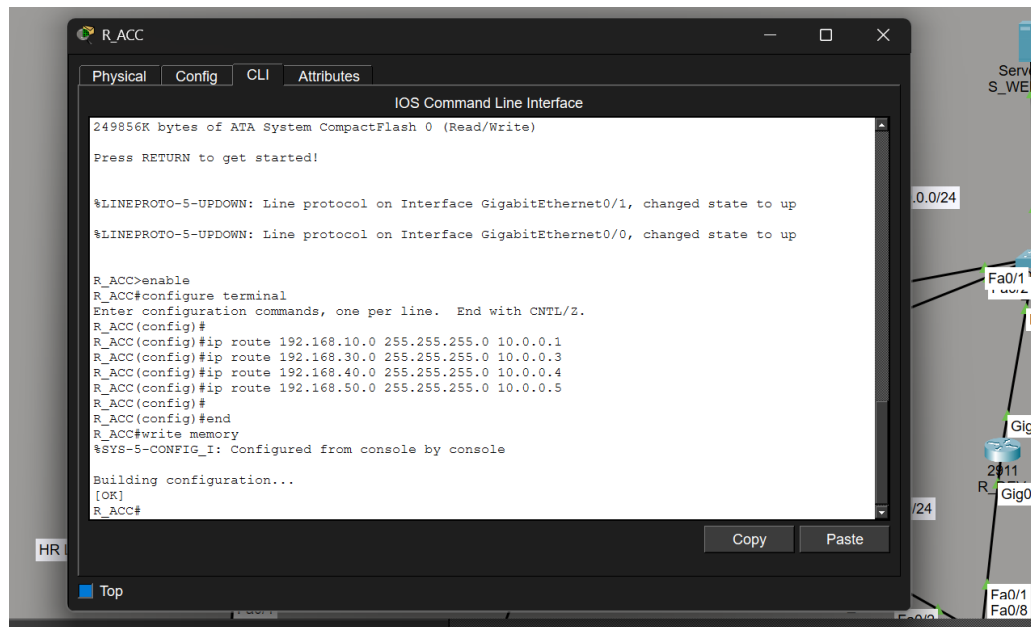


Figure 11: Static route configuration on the HR router

3.4 Network without ACL

3.4.1 HTTP and FTP server

One server was used for both web and file services. The following services were enabled:

- HTTP,
- HTTPS,
- FTP.

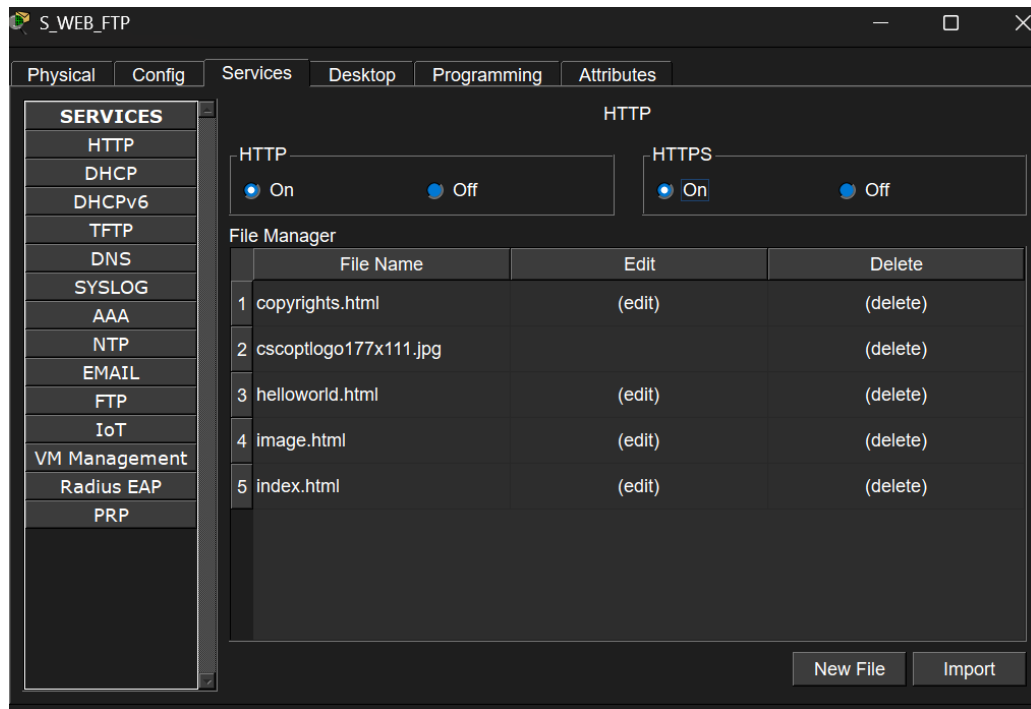


Figure 12: HTTP and HTTPS services enabled on the server

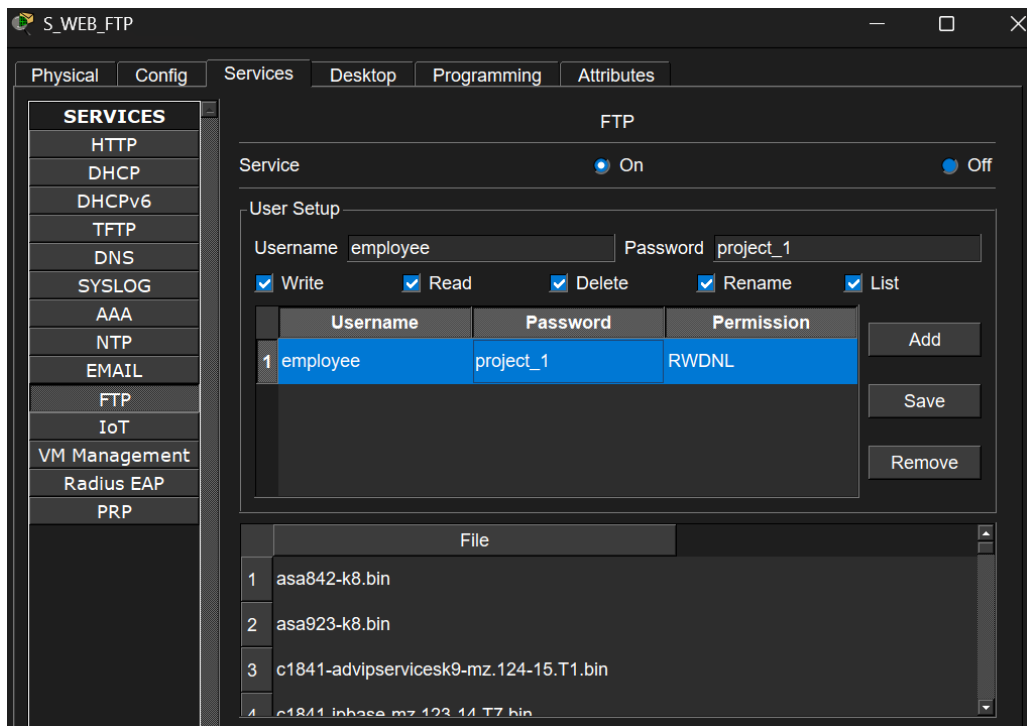


Figure 13: FTP service enabled on the server

The default FTP account was not used. Instead, a new account was created:

- username: `employee`,
- password: `project_1`.

Changing default credentials is a basic security practice. Even in a simulation, it is better not to leave default usernames and passwords.

3.4.2 Tests without ACL

Before applying ACLs, the network was tested to check whether addressing and routing were configured correctly. At this stage, no traffic restrictions were active, so communication between subnets was expected to work.

Example test from the HR PC:

```
C:\>ping 192.168.10.1

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=9ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Ping to the server:

```
C:\>ping 10.0.0.10
```



```
Request timed out.  
Reply from 10.0.0.10: bytes=32 time=19ms TTL=127  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127
```

The first packet was lost, but after that the connection worked. This is acceptable in Packet Tracer, especially during the first communication attempt, because ARP resolution may still be in progress.

Ping to another department before ACL:

```
C:\>ping 192.168.20.10  
  
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126  
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126  
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126  
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
```

This confirms that interdepartment routing was working before ACL rules were applied. The HTTP service was tested using a web browser.

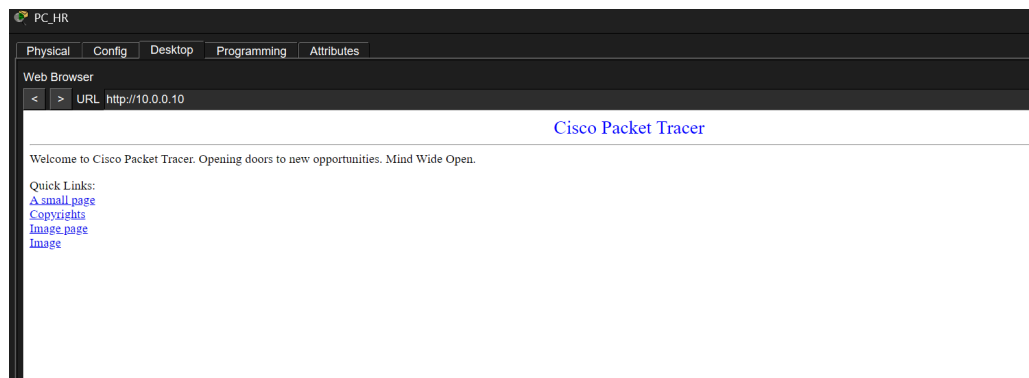


Figure 14: Successful HTTP access to the intranet server

The FTP service was also tested.

```
C:\>ftp 10.0.0.10  
Trying to connect...10.0.0.10  
Connected to 10.0.0.10  
220- Welcome to PT Ftp server  
Username: employee  
331- Username ok, need password  
Password: *****  
230- Logged in  
(passive mode On)  
  
ftp>dir  
Listing /ftp directory from 10.0.0.10:  
0 : asa842-k8.bin  
1 : asa923-k8.bin  
2 : c1841-advipservicesk9-mz.124-15.T1.bin  
...
```

```
ftp>quit
221- Service closing control connection.
```

At this point, addressing, routing, HTTP and FTP services were working correctly. The next step was to add access restrictions.

3.5 Access Control Lists

The required access model is shown in Figure 15. The President and the Secretary should have full access, while ordinary departments should only access their own subnet and the intranet server.

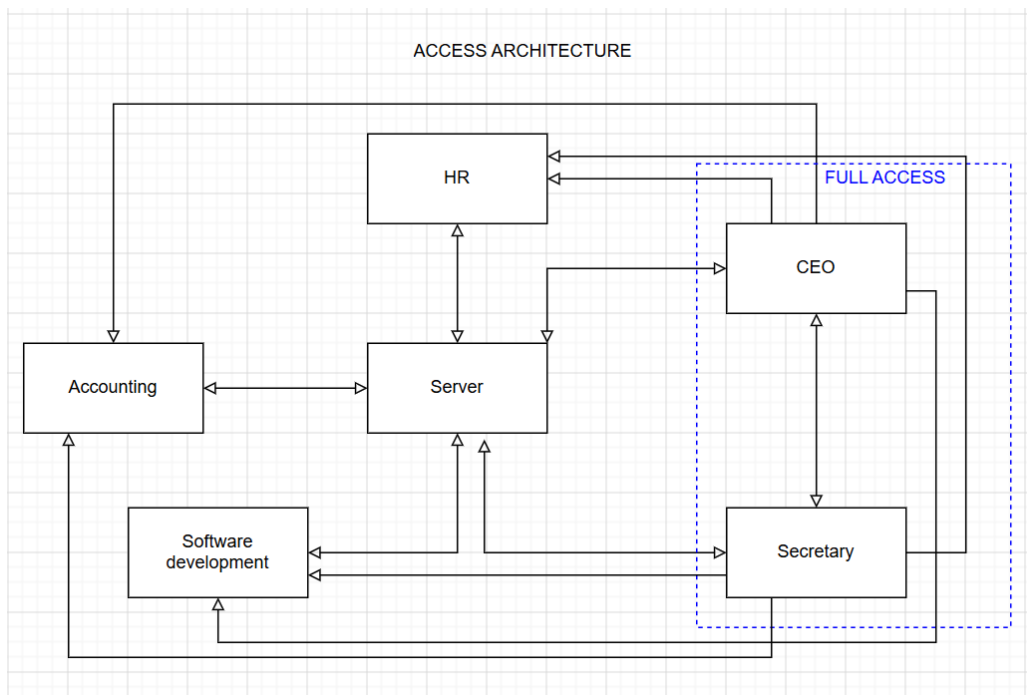


Figure 15: Access policy diagram prepared in diagrams.net

ACLs were applied only on the routers of ordinary departments:

- R_HR,
- R_ACC,
- R_DEV.

No restrictive ACL was applied on R_CEO and R_SEC, because the President and the Secretary are allowed to access all computers.

The ACLs were applied inbound on the `GigabitEthernet0/0` interfaces. In this way, traffic coming from a local department LAN is checked before it is routed to another network.

One additional rule was needed for ICMP replies. For example, if the President pings the HR PC, the HR PC must be able to send an ICMP `echo-reply` back to the President. Without this rule, the request would reach the HR PC, but the response would be blocked.

3.5.1 ACL configuration for HR

```
enable
configure terminal

access-list 110 permit ip 192.168.10.0 0.0.0.255 host 10.0.0.10
access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-
    reply
access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255 echo-
    reply
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 110 permit ip any any

interface gigabitEthernet0/0
ip access-group 110 in
exit

end
write memory
```

3.5.2 ACL configuration for Accounting

```
enable
configure terminal

access-list 120 permit ip 192.168.20.0 0.0.0.255 host 10.0.0.10
access-list 120 permit icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-
    reply
access-list 120 permit icmp 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255 echo-
    reply
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 120 permit ip any any

interface gigabitEthernet0/0
ip access-group 120 in
exit

end
write memory
```

3.5.3 ACL configuration for Development

```
enable
```

```

configure terminal

access-list 130 permit ip 192.168.30.0 0.0.0.255 host 10.0.0.10
access-list 130 permit icmp 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-
    reply
access-list 130 permit icmp 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255 echo-
    reply
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 130 permit ip any any

interface gigabitEthernet0/0
ip access-group 130 in
exit

end
write memory

```

3.5.4 Explanation of selected ACL commands

The following command allows the HR subnet to access the internal server:

```
access-list 110 permit ip 192.168.10.0 0.0.0.255 host 10.0.0.10
```

This means that any host from 192.168.10.0/24 can communicate with the server 10.0.0.10. As a result, HR users can use services such as HTTP, HTTPS and FTP.

The following commands allow ICMP replies from HR to the President and Secretary networks:

```

access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-
    reply
access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255 echo-
    reply

```

These commands are needed because the President and the Secretary must be able to ping HR computers. The HR computer must therefore be allowed to send an ICMP reply back.

The following commands block HR from starting communication with other departments:

```

access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255

```

The last explicit rule is:

```
access-list 110 permit ip any any
```

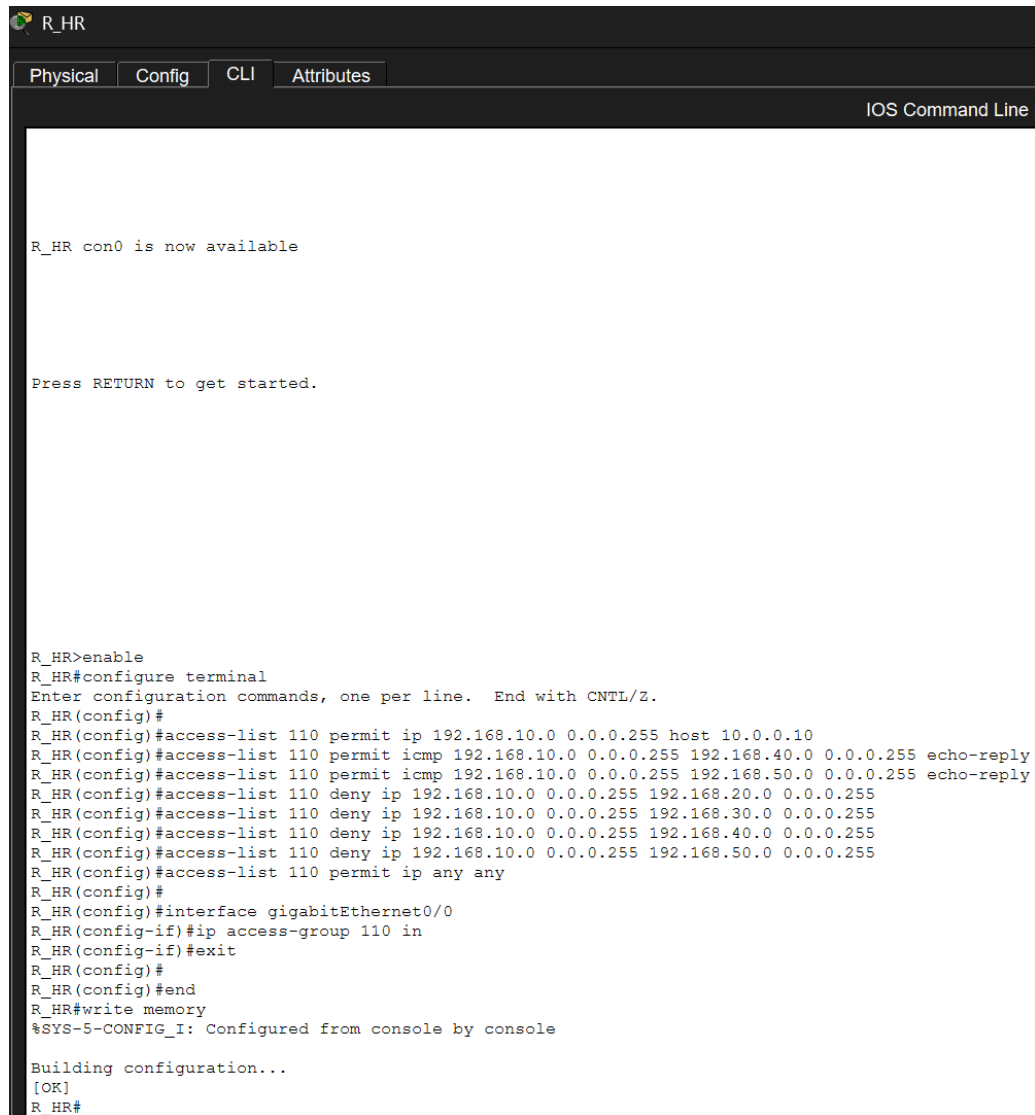
This rule allows all traffic that was not matched by previous ACL entries. It is important because Cisco ACLs have an implicit **deny ip any any** rule at the end. Without the final permit rule, more traffic could be blocked than intended.

Cisco ACLs use wildcard masks. For a /24 subnet, the subnet mask is 255.255.255.0, while the wildcard mask is 0.0.0.255. In a wildcard mask, zero means that the value must match, while 255 means that the value may vary.

After creating the ACL, it has to be assigned to the correct interface:

```
interface gigabitEthernet0/0
ip access-group 110 in
```

This command applies ACL 110 inbound on the HR router interface connected to the HR LAN.



```
R_HR
Physical Config CLI Attributes
IOS Command Line Interface

R_HR con0 is now available

Press RETURN to get started.

R_HR>enable
R_HR#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R_HR(config)#
R_HR(config)#access-list 110 permit ip 192.168.10.0 0.0.0.255 host 10.0.0.10
R_HR(config)#access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-reply
R_HR(config)#access-list 110 permit icmp 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255 echo-reply
R_HR(config)#access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R_HR(config)#access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
R_HR(config)#access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
R_HR(config)#access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
R_HR(config)#access-list 110 permit ip any any
R_HR(config)#
R_HR(config)#interface gigabitEthernet0/0
R_HR(config-if)#ip access-group 110 in
R_HR(config-if)#exit
R_HR(config)#
R_HR(config)#end
R_HR#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R_HR#
```

Figure 16: ACL assigned to the HR router

3.5.5 Connectivity tests after ACL

After ACL configuration, the network was tested again. The goal was to check whether allowed traffic works and forbidden traffic is blocked.

Test	Expected result	Actual result
PC_HR → Server	Allowed	Success
PC_HR → PC_ACC1	Denied	Failed
PC_HR → PC_DEV1	Denied	Failed
PC_ACC1 → Server	Allowed	Success
PC_ACC1 → PC_DEV1	Denied	Failed
PC_DEV1 → Server	Allowed	Success
PC_DEV1 → PC_HR	Denied	Failed
PC_CEO → PC_HR	Allowed	Success
PC_CEO → PC_ACC1	Allowed	Success
PC_CEO → PC_DEV1	Allowed	Success
PC_SEC → PC_HR	Allowed	Success
PC_SEC → PC_DEV1	Allowed	Success
PC_SEC → Server	Allowed	Success

Table 7: Connectivity tests after ACL implementation

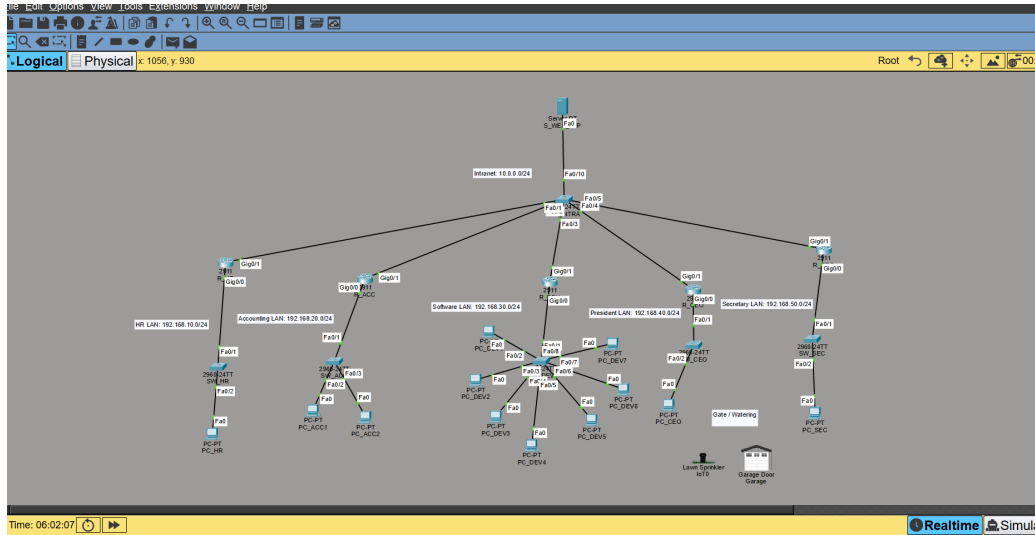


Figure 17: Verification of network behaviour after ACL implementation

The tests confirmed that the network works according to the requirements. Ordinary departments can access the server, but cannot access other departments. The President and the Secretary can access all tested computers.

3.6 IoT devices

The last part of the project was to add two IoT devices:

- Garage Door,
- Lawn Sprinkler.

These devices represent the company gate and the lawn watering system. According to the requirements, the Secretary must be able to access them from her computer.

The IoT devices were connected to the Secretary switch and placed in the Secretary LAN. This solution is simple and fits the access policy, because the Secretary is in the same subnet as the IoT devices.

During configuration, it was necessary to replace or add a network module in the IoT devices to make a FastEthernet port available.

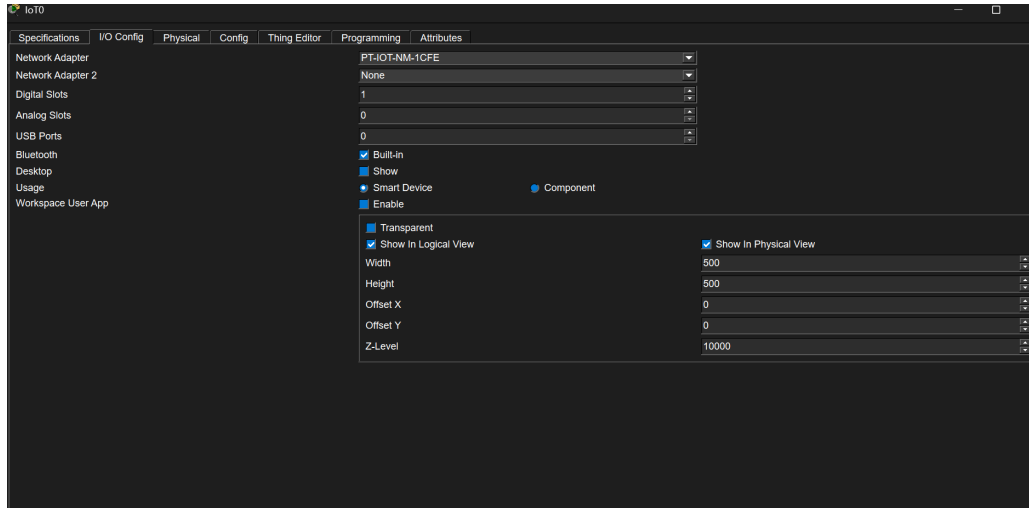


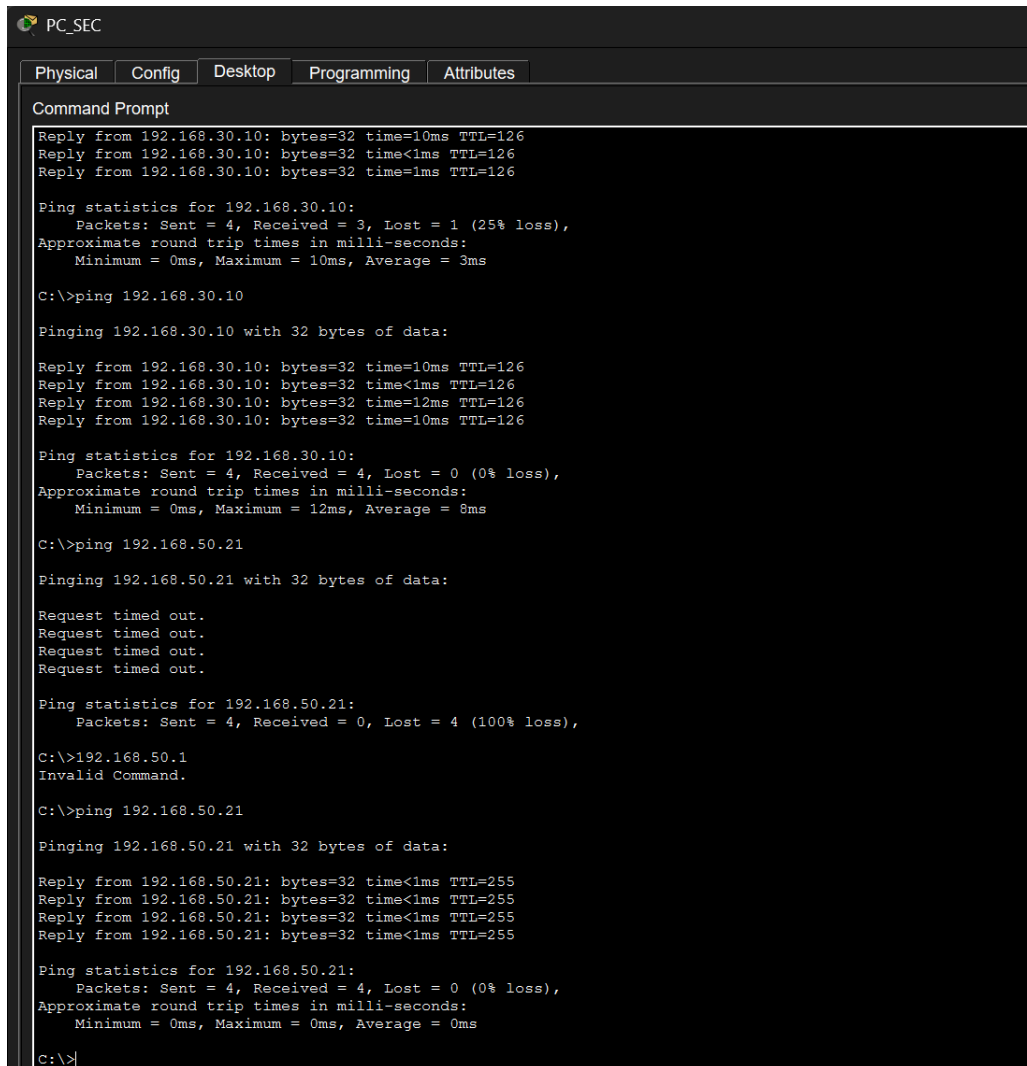
Figure 18: Replacing the network module in an IoT device

The IoT devices were configured as shown in Table 8.

Device	IP address	Subnet mask	Default gateway
Garage Door	192.168.50.20	255.255.255.0	192.168.50.1
Lawn Sprinkler	192.168.50.21	255.255.255.0	192.168.50.1

Table 8: IoT device addressing

The Secretary PC was able to reach both IoT devices.



```
PC_SEC
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.30.10: bytes=32 time=10ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=10ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=12ms TTL=126
Reply from 192.168.30.10: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 8ms

C:\>ping 192.168.50.21

Pinging 192.168.50.21 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.50.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>192.168.50.1
Invalid Command.

C:\>ping 192.168.50.21

Pinging 192.168.50.21 with 32 bytes of data:

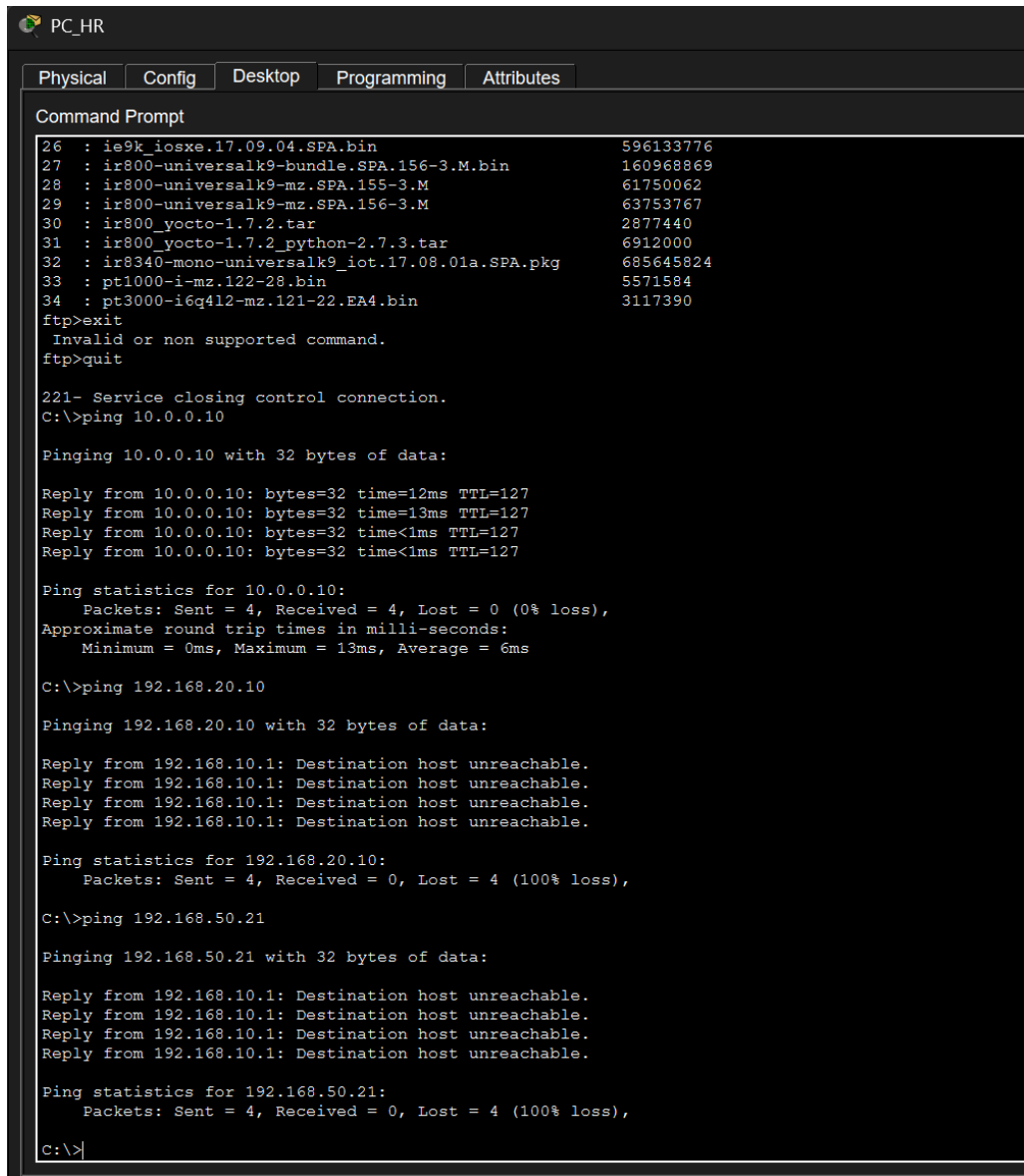
Reply from 192.168.50.21: bytes=32 time<1ms TTL=255
Reply from 192.168.50.21: bytes=32 time<1ms TTL=255
Reply from 192.168.50.21: bytes=32 time<1ms TTL=255
Reply from 192.168.50.21: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.50.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 19: Successful ping test from the Secretary PC to IoT devices

The HR PC could not reach the IoT devices, which is the expected result because HR traffic to the Secretary LAN is blocked by ACL.



The screenshot shows a Windows window titled "PC_HR" with a tabbed interface. The "Physical" tab is selected. Below the tabs is a "Command Prompt" window. The command prompt shows a list of files and their MD5 hashes, followed by an attempt to exit the FTP session, which fails with the message "Invalid or non supported command." The user then enters "ftp>quit", and the connection closes. The user then runs a series of ping commands to various IP addresses: 10.0.0.10, 192.168.20.10, and 192.168.50.21. The first ping to 10.0.0.10 is successful, showing four replies and statistics. The subsequent pings to 192.168.20.10 and 192.168.50.21 are unsuccessful, showing four replies each with "Destination host unreachable" and 100% packet loss.

```
PC_HR
Physical Config Desktop Programming Attributes
Command Prompt
26 : ie9k_iosxe.17.09.04.SPA.bin 596133776
27 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ir800-universalk9-mz.SPA.155-3.M 61750062
29 : ir800-universalk9-mz.SPA.156-3.M 63753767
30 : ir800_yocto-1.7.2.tar 2877440
31 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin 5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin 3117390
ftp>exit
Invalid or non supported command.
ftp>quit
221- Service closing control connection.
C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=12ms TTL=127
Reply from 10.0.0.10: bytes=32 time=13ms TTL=127
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 6ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.50.21

Pinging 192.168.50.21 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.50.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Figure 20: Unsuccessful ping test from HR PC to IoT devices – expected behaviour

The screenshot below shows an example of the interactive IoT features: the garage door is open and the lawn sprinkler is active in front of the office.

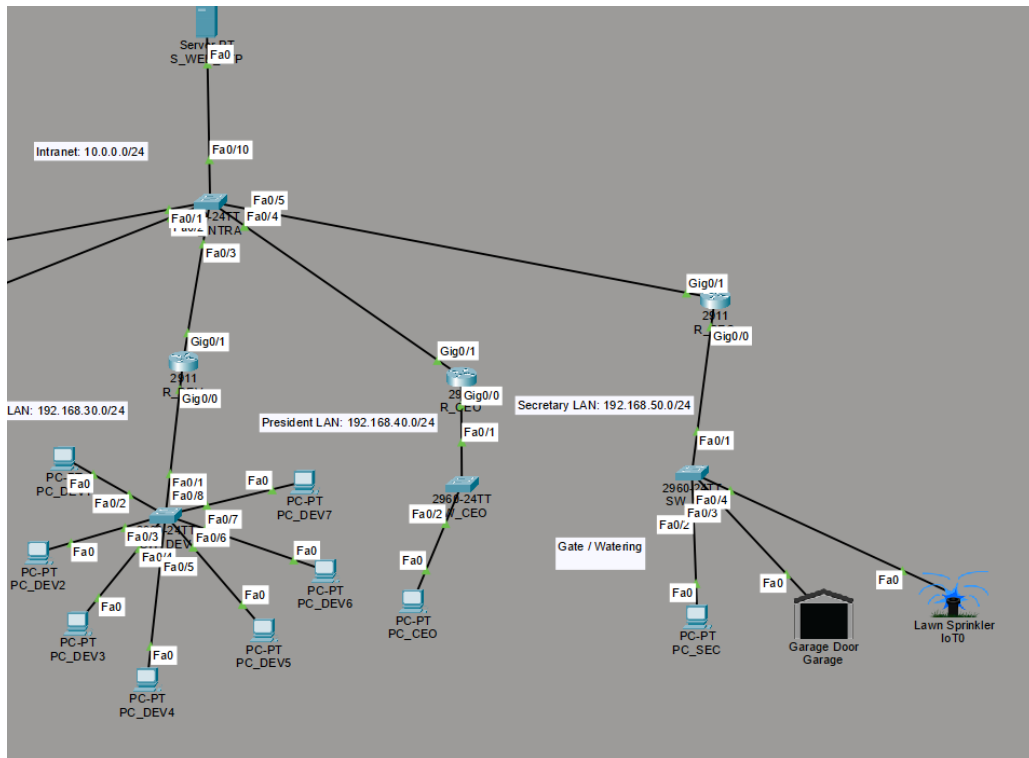


Figure 21: Interactive IoT features enabled

The final topology is shown in Figure 22.

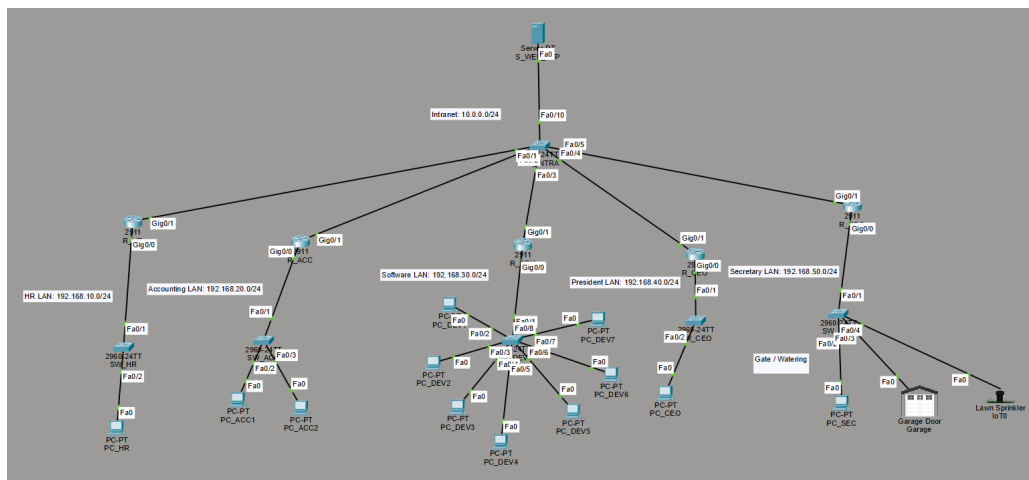


Figure 22: Final logical topology

4 Cost estimate

The project was prepared in Cisco Packet Tracer, so no real devices were purchased. However, an approximate cost estimate was prepared to show how much a similar configuration could cost in a real environment.

According to the project requirements, the cost estimate includes only network devices and installation materials. Office computers are shown separately only for comparison and are not included in the required project cost.

Item	Quantity	Estimated unit price [PLN]	Estimated total [PLN]
Cisco 2911 access router	5	300	1500
Cisco 2960 switch	6	250	1500
Server for HTTP/FTP services	1	2000	2000
IoT device: gate controller	1	300	300
IoT device: lawn watering controller	1	300	300
Ethernet patch cables	25	10	250
Basic installation materials	1	500	500
Office computer	12	1800	21600
Estimated total without office computers			6350
Estimated total with office computers			27950

Table 9: Approximate cost estimate of devices and materials

The prices are approximate and were used for educational purposes only. The estimates were based on prices found on Allegro and ITPrice. In a real company, the final price could be different depending on the selected hardware, supplier and licences. The estimate does not include labour costs, rack cabinets, UPS devices, patch panels or structured cabling installation.

5 Conclusions

5.1 Observations

During the project we had a few small configuration problems. At the beginning, some PCs could ping only their own router, but they could not reach the server. The problem was caused by missing static routes on some routers. After adding routes to the other departmental networks, communication started to work correctly.

Another problem appeared during server testing. The PCs were configured correctly, but the server did not answer from other subnets. After checking the server settings, we noticed that the default gateway was missing. We set the gateway to 10.0.0.1, which is the address of the HR router in the intranet network, and after that the server was reachable.

We also had to correct the ACL configuration. In the first version, the President and the Secretary could send ping requests to ordinary departments, but the replies were blocked. To fix this, we added rules allowing ICMP `echo-reply` packets from HR, Accounting and Development networks to the President and Secretary networks.

The project showed that in Packet Tracer even small mistakes, such as a missing gateway or one missing static route, can block communication between networks.

5.2 Achievements

The final Packet Tracer project works according to the main requirements. We created separate LANs for HR, Accounting, Software Development, the President office and the Secretary office. Each LAN has its own router and all routers are connected to one intranet network with the server.

The web and FTP server can be reached from all employee computers. HR, Accounting and Development computers cannot access computers from other departments after ACLs are applied. At the same time, the President and the Secretary can access all tested computers.

The Secretary can also reach the IoT devices connected in her LAN. The Garage Door and Lawn Sprinkler were configured with static IP addresses and tested using ping from the Secretary PC.

The final tests confirmed that addressing, static routing, server access, ACL rules and IoT connectivity were configured correctly.

5.3 Development potential

The project could be developed further by adding DHCP service. In the current version, all IP addresses were entered manually. This is acceptable for a small Packet Tracer project, but in a real company it would be easier to assign addresses automatically.

Another small improvement could be adding descriptions to all switch ports and router interfaces. This would make troubleshooting easier, especially when the topology becomes larger.

Based on the first laboratory exercise, another possible extension would be to add an IP camera. The camera could include a simple interaction mechanism in Packet Tracer, where it turns on when the user hovers over it with the mouse. This would simulate automatic motion detection. An extension could improve security by monitoring the area around the gate and the lawn sprinkler system.

The current solution is enough for the project requirements, but DHCP would make the network easier to manage if more computers were added later.

References

- [1] Cisco Networking Academy, *Cisco Packet Tracer learning materials and tutorials*. Available at: <https://www.netacad.com/courses/packet-tracer>
- [2] Cisco, *Access Control Lists configuration examples and documentation*. Available at: <https://www.cisco.com/>
- [3] Cisco, *Static routing configuration examples and documentation*. Available at: <https://www.cisco.com/>
- [4] Cisco Packet Tracer, *Built-in device configuration help and simulation tools*.
- [5] ITPrice, *Estimated network device prices used for comparison*. Available at: <https://itprice.com/>